

# User Funnel Drop-off Analysis

Data Analyst Assessment – Fobes Skills

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Python | Pandas | Matplotlib

## 1. Project Overview

This project analyzes user behavior across a multi-stage conversion funnel to understand how users progress from visiting a website to completing a purchase. The analysis focuses on measuring the number of users reaching each funnel stage, calculating stage-to-stage conversion and drop-off rates, and identifying the stage where the highest proportion of users are lost.

The findings are used to identify potential areas of friction in the user journey and provide business recommendations for improving conversion.

## 2. Business Problem

Users may abandon a website or application at different stages of the conversion journey.

Understanding where users drop off helps businesses identify potential friction points and prioritize areas for improvement.

This analysis aims to answer:

- How many users reach each stage of the funnel?
- What percentage of users progress from one stage to the next?
- Where does the largest drop-off occur?
- How much time do users take to move between consecutive stages?
- What actions could help reduce user drop-off?

## 3. Objectives

- Analyze user movement through the conversion funnel.
- Calculate the number of unique users at each stage.
- Calculate stage-to-stage conversion rates.
- Calculate drop-off rates between consecutive stages.
- Identify the stage with the largest drop-off.
- Analyze the average time taken between consecutive funnel stages.
- Provide actionable business recommendations based on the findings.

## 4. Dataset

The dataset contains event-level records representing user activity throughout the funnel.

| Column           | Description                                    |
|------------------|------------------------------------------------|
| <b>user_id</b>   | Unique identifier for each user                |
| <b>step</b>      | Funnel stage associated with the event         |
| <b>timestamp</b> | Date and time at which the user event occurred |

The funnel consists of five stages: **Visited Site, Signup Started, Details Filled, Email Verified, Purchase Completed.**

The analysis identifies 200 unique users in the dataset.

## 5. Tools & Technologies

- **Python** — analysis and data processing
- **Pandas** — data manipulation and calculations
- **Matplotlib** — data visualization
- **Google Colab** — development environment

## 6. Methodology

### Data Exploration

The dataset was inspected to understand its structure, columns, data types, missing values, number of records, unique users, and available funnel stages.

### Unique User Calculation

The number of unique users reaching each funnel stage was calculated using `user_id`.

### Conversion Rate Analysis

Stage-to-stage conversion rates were calculated by comparing the number of unique users at the current stage with the number of unique users at the previous stage.

### Drop-off Rate Analysis

The drop-off rate was calculated as the percentage of users who did not progress from the previous funnel stage to the current stage.

### Largest Drop-off Identification

The drop-off rates were compared to identify the stage with the highest user loss.

### Time Between Stages

Timestamps were converted into datetime format and users were ordered by timestamp. Valid transitions between consecutive funnel stages were used to calculate the average time taken between stages.

## 7. Funnel Analysis

The analysis evaluates user progression through five stages of the conversion journey:

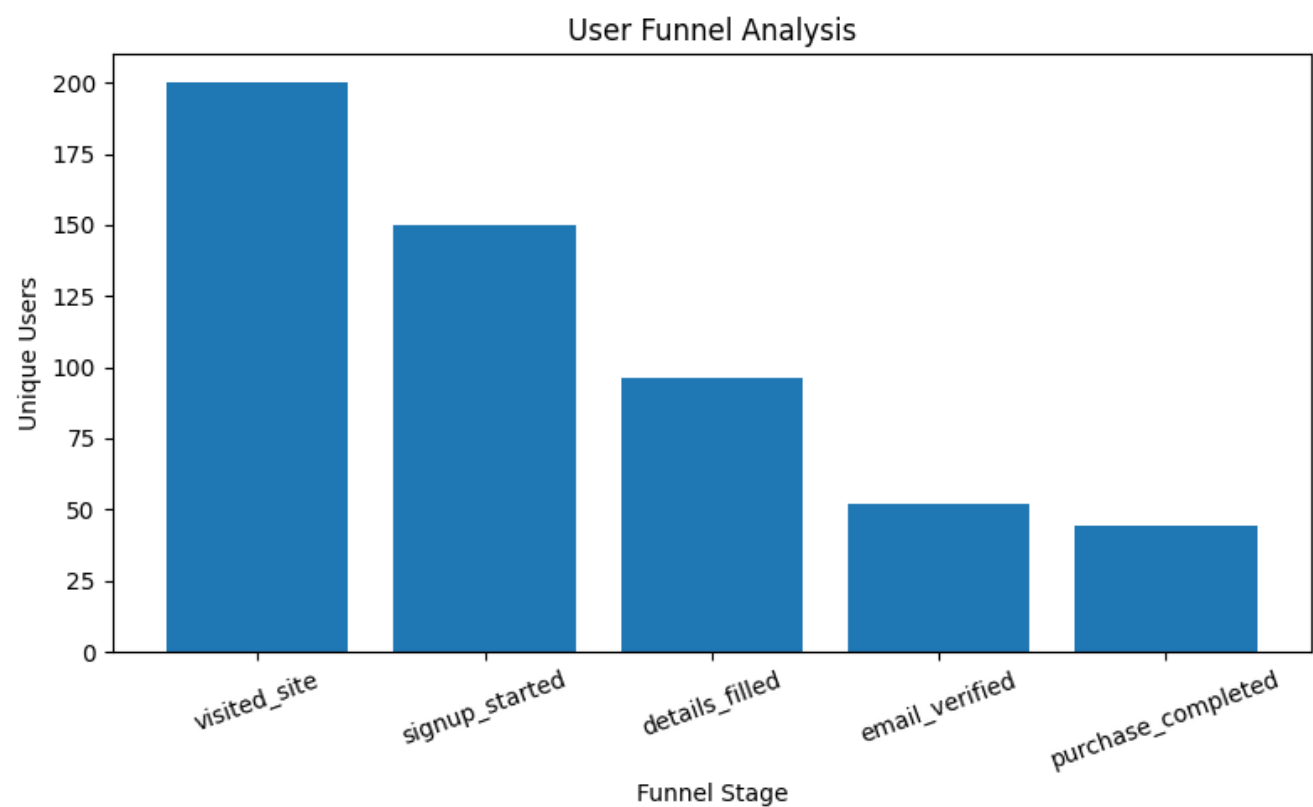


Figure 1: Number of unique users at each funnel stage

| Stage | Funnel Step        |
|-------|--------------------|
| 1     | Visited Site       |
| 2     | Signup Started     |
| 3     | Details Filled     |
| 4     | Email Verified     |
| 5     | Purchase Completed |

### Major Drop-off

The largest drop-off occurs between **Details Filled** and **Email Verified**. The analysis identifies a **45.83% drop-off** at this transition, making the email verification stage the primary area for investigation and improvement.

## 8. Key Insights

### User conversion decreases throughout the funnel

The number of users reaching each subsequent stage decreases as users progress through the conversion journey.

### Email verification is the major drop-off point

The highest drop-off occurs after users complete their details but before they verify their email. The observed drop-off is 45.83%.

### **Email verification may represent a friction point**

The high loss of users at this stage suggests that the verification process may be creating friction in the user journey. This stage should therefore be prioritized for further investigation.

## **9. Business Recommendations**

### **Simplify the verification process**

Reduce unnecessary steps involved in email verification so that users can complete the process more easily.

### **Enable one-click verification**

A direct verification link can reduce the effort required from users to complete the verification step.

### **Send reminder emails**

Users who have not completed verification could receive reminder emails to encourage them to return and finish the process.

### **Clearly communicate the purpose of verification**

Providing a short explanation of why email verification is required may increase user confidence and encourage completion.

### **Investigate verification-related issues**

The business could further investigate whether users are experiencing delayed emails, difficulty locating verification messages, or unclear instructions.

## **10. Conclusion**

This funnel analysis evaluates user progression across five stages of a conversion journey and identifies where the largest proportion of users are lost.

The analysis highlights the transition from **Details Filled to Email Verified** as the major drop-off point, with a **45.83%** drop-off rate.

Improving the email verification experience could therefore be an important opportunity for reducing abandonment and improving overall funnel conversion.

This project demonstrates the use of **Python, Pandas, and Matplotlib** to transform event-level user data into actionable business insights and recommendations.